

FCC 认证测试设置

Pick worst case

1, Band Edges Emission

蓝牙频带外 100kHz 带宽内的信号必须低于最接近通道信号最大值 20dB, 针对最低 (2.400GHz)和最高 channel(2.4835GHz)。

TEST SETTINGS:

- (1) 最大功率发射; EUT 连续发射 or maxhold&peak
- (2) RBW=100KHz($\geq 1\%$ span=10MHz), VBW=300KHz($\geq$ RBW)
- (3) 跳频模式打开, 重复(1) (2)

2, Spurious Emission

TEST SETTINGS:

- (1) 线损补偿后测试
- (2) 最大功率发射; EUT 连续发射 or maxhold&peak
- (3) RBW=100KHz, VBW=300KHz, span 1GHz~26GHz, 所有的杂散必须低于最高值 20dB
(CE 认证 30M~1G -36dBm; 1G~12.75G -30dBm; 1.8G~1.9G & 5.15G~5.3G -47dBm)
- (4) 1M 低中高 channel

3, 20dB BW

- (1) 最大功率发射; EUT 连续发射 or maxhold&peak
- (2) Span = 2~3 倍的 20dB BW (20~30MHz)
- (3) RBW $\geq 1\%$ 20dB BW, VBW $\geq$ RBW
- (4) 低中高 channel

4, Number of Hopping Channel

- (1) 最大功率发射; EUT 连续发射 or maxhold&peak
- (2) 跳频模式
- (3) Span: 2.400GHz~2.4835GHz; RBW $\geq 1\%$ of the span; VBW $\geq$ RBW (RBW=100K/1M, VBW=300K/1M)

5, Hopping Channel Separation Measurement

Channel 之间须有>25KHz 的间隔

- (1) 最大功率发射; EUT 连续发射 or maxhold&peak
- (2) 跳频模式
- (3) Span = wide enough to capture the peaks of two adjacent channels; RBW $\geq 1\%$ of the span; VBW $\geq$ RBW (span=2M, RBW=30K, VBW=100K)
- (4) 低中高 channel

6, Peak Output Power

- (1) 最大功率发射; EUT 连续发射 or maxhold&peak
- (2) 线损补偿
- (3) 低中高 channel

7, Dwell Time

DH1 Dwell time = $t \cdot (1.6/2/79) \cdot 31.6$

DH3 Dwell time = $t \cdot (1.6/4/79) \cdot 31.6$

DH5 Dwell time = $t \cdot (1.6/6/79) \cdot 31.6$ (limit: 400ms)

(1) 最大功率发射; EUT 连续发射 or maxhold&peak

(2) 跳频模式

(3) Span = zero span, centered on a hopping channel; RBW=1MHz, VBW $\geq$ RBW; sweep time is more than once pulse time

8, RX Spurious Emission

CE 有要求

The spurious emission for a Bluetooth receiver shall not be more than:

Frequency Band	Requirement
30 MHz - 1 GHz	-57 dBm
1 GHz – 12.75 GHz	-47 dBm

Table 4.3: Out-of-band spurious emission

The measured power should be measured in a 100 kHz bandwidth.